International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Further Practice Problems with Solutions

Sequences, series, and the real line

Definitions and facts assumed

The solutions below are self-contained. Every result used in a proof is stated where it is used, and no question refers to any other. The following definitions and facts are assumed throughout and are not reproved.

Definitions.

(D1) A sequence $(a_n)_{n=m}^{\infty}$ of reals *converges to* a real L iff for every real $\varepsilon > 0$ there is $N \geq m$ with $|a_n - L| \leq \varepsilon$ for all $n \geq N$; we write $L = \lim_{n\to\infty} a_n$. A sequence converging to no real number *diverges*.

(D2) $(a_n)_{n=m}^{\infty}$ is *bounded* iff there is a real $M \geq 0$ with $|a_n| \leq M$ for all $n \geq m$.

(D3) $(a_n)_{n=m}^{\infty}$ is *increasing* iff $a_n \leq a_{n+1}$ for all $n \geq m$, and *decreasing* iff $a_n \geq a_{n+1}$ for all $n \geq m$; it is *monotone* iff it is one or the other.

(D4) $(b_j)_{j=0}^{\infty}$ is a *subsequence* of $(a_n)_{n=0}^{\infty}$ iff there are indices $n_0 < n_1 < n_2 < \cdots$ in $\mathbb{N}$ with $b_j = a_{n_j}$ for every j.

(D5) A real L is a *limit point* of $(a_n)_{n=m}^{\infty}$ iff for every $\varepsilon > 0$ and every $N \geq m$ there is $n \geq N$ with $|a_n - L| \leq \varepsilon$.

(D6) For $E \subseteq \mathbb{R}$ non-empty and bounded above, $\sup(E)$ is its least upper bound; $\sup(E) := +\infty$ if E is non-empty and unbounded above; $\sup(\emptyset) := -\infty$; and $\inf(E) := -\sup(-E)$, where $-E := \{-x : x \in E\}$. For a sequence $(a_n)_{n=m}^{\infty}$ and $N \geq m$ put $a_N^+ := \sup\{a_n : n \geq N\}$ and $a_N^- := \inf\{a_n : n \geq N\}$; then $\limsup_{n\to\infty} a_n := \inf\{a_N^+ : N \geq m\}$ and $\liminf_{n\to\infty} a_n := \sup\{a_N^- : N \geq m\}$.

(D7) Given reals $a_m, a_{m+1}, \ldots$, the series $\sum_{n=m}^{\infty} a_n$ has *partial sums* $S_N := \sum_{n=m}^{N} a_n$. If $(S_N)_{N=m}^{\infty}$ converges to a real L, the series is *convergent* with *sum* L, written $L = \sum_{n=m}^{\infty} a_n$; otherwise it is *divergent*.

(D8) $\sum_{n=m}^{\infty} a_n$ is *absolutely convergent* iff $\sum_{n=m}^{\infty} |a_n|$ converges, and *conditionally convergent* iff it converges but $\sum_{n=m}^{\infty} |a_n|$ does not.

Facts.

(F1) $|a+b| \le |a| + |b|$, $|ab| = |a|\,|b|$, $|-a| = |a|$, and $|a| = 0$ iff $a = 0$.

(F2) *(Archimedean property.)* For every real x there is a positive integer N with $x \le N$.

(F3) *(Limit laws.)* If $a_n \to L$ and $b_n \to M$ then $a_n + b_n \to L + M$, $a_n b_n \to LM$ and $ca_n \to cL$ for every real c; and if $M \neq 0$ and every $b_n \neq 0$, then $a_n / b_n \to L/M$.

(F4) Limits preserve non-strict inequalities. If $a_n \to L$ and $a_n \ge c$ for all n then $L \ge c$, and likewise with $\le$.

(F5) *(Squeeze test.)* If $b_n \le a_n \le c_n$ for all n and $b_n \to L$ and $c_n \to L$, then $a_n \to L$.

(F6) A sequence of reals converges if and only if it is Cauchy, meaning that for every $\varepsilon > 0$ there is N with $|a_j - a_k| \le \varepsilon$ for all $j, k \ge N$.

(F7) *(Monotone convergence.)* An increasing sequence that is bounded above converges, to the supremum of its set of values; a decreasing sequence bounded below converges, to the infimum of its set of values.

(F8) *(Suprema and infima.)* For $E \subseteq \mathbb{R}$ non-empty and bounded above, $\sup(E)$ is a real number, $x \le \sup(E)$ for every $x \in E$, $\sup(E) \le M$ for every upper bound M, and for every $\varepsilon > 0$ there is $y \in E$ with $y > \sup(E) - \varepsilon$. Symmetrically for infima and sets bounded below.

(F9) Every subsequence of a convergent sequence converges to the same limit, and a sequence cannot converge to two distinct limits.

(F10) $\lim_{n\to\infty} 1/n^q = 0$ for every rational $q > 0$.

(F11) For a real $x \ge 0$ and an integer $n \ge 1$, $x^{1/n}$ is the unique non-negative real whose nth power is x; and $x > y \ge 0$ iff $x^{1/n} > y^{1/n}$.

(F12) For every real x there is a unique integer $\lfloor x \rfloor$ with $\lfloor x \rfloor \le x < \lfloor x \rfloor + 1$.

(F13) *(Geometric series.)* For a real $r \neq 1$ and every $N \ge 0$, $\sum_{k=0}^{N} r^k = (1 - r^{N+1})/(1 - r)$. If $0 \le r < 1$ the series $\sum_{k=0}^{\infty} r^k$ converges to $1/(1-r)$; if $r \ge 1$ it diverges.

(F14) A series whose terms are all non-negative converges if and only if its partial sums are bounded above.

(F15) *(Comparison test.)* If $0 \le a_n \le b_n$ for all n and $\sum b_n$ converges, then $\sum a_n$ converges; equivalently, if $\sum a_n$ diverges then so does $\sum b_n$.

(F16) *(Alternating series test.)* If $(a_n)_{n=m}^{\infty}$ is non-negative and decreasing, then $\sum_{n=m}^{\infty}(-1)^n a_n$ converges if and only if $a_n \to 0$.

(F17) *(Zero test.)* If $\sum_{n=m}^{\infty} a_n$ converges then $a_n \to 0$.

Each question is followed by a short note on what makes it worth doing, and then by a solution.

Part A. Sequences

Question (1). Define a sequence by

$$a_1 := 2, \qquad a_{n+1} := \frac{1}{2}\Big(a_n + \frac{2}{a_n}\Big) \quad (n \ge 1).$$

(a) Show that $a_n > 0$ and $a_n^2 \ge 2$ for every $n \ge 1$.

(b) Show that (a_n) is decreasing, and deduce that it converges.

(c) Show that its limit is $2^{1/2}$.

(d) The first three terms are

$$2, \quad \frac{3}{2}, \quad \frac{17}{12},$$

whose squares exceed 2 by 2, by 2.5×10^{-1} and by 6.9×10^{-3} respectively. Prove the identity

$$a_{n+1}^2 - 2 \;=\; \frac{(a_n^2 - 2)^2}{4a_n^2}$$

and explain what it says about the speed of the approximation.

Why this is interesting. Most sequences in a first course are given by a closed formula, and the limit is read off. This one is given only by a rule for passing from one term to the next, so the limit has to be obtained indirectly, by first proving convergence and then letting $n \to \infty$ inside the recursion. That two-step strategy is the standard one for recursively defined sequences. The sequence is also the Babylonian method for

extracting square roots, in use for about four thousand years and still what a computer does, and part (d) explains why a calculator produces a square root so quickly, since the number of correct digits roughly doubles at each step.

(a) We show by induction that $a_n > 0$; then the formula makes sense at every stage. Certainly $a_1 = 2 > 0$; and if $a_n > 0$ then $2/a_n > 0$ and hence $a_{n+1} = \frac{1}{2}(a_n + 2/a_n) > 0$.

For the second claim, first note the identity

$$a_{n+1}^2 - 2 \;=\; \Big(\frac{a_n^2+2}{2a_n}\Big)^2 - 2 \;=\; \frac{a_n^4 + 4a_n^2 + 4 - 8a_n^2}{4a_n^2} \;=\; \frac{(a_n^2-2)^2}{4a_n^2}, \qquad (*)$$

which is the identity of part (d). Its right-hand side is a square divided by a positive number, hence ≥ 0. So $a_{n+1}^2 \geq 2$ for every $n \geq 1$; and $a_1^2 = 4 \geq 2$ as well.

(b) For $n \geq 1$,

$$a_n - a_{n+1} \;=\; a_n - \frac{1}{2}\Big(a_n + \frac{2}{a_n}\Big) \;=\; \frac{1}{2}\Big(a_n - \frac{2}{a_n}\Big) \;=\; \frac{a_n^2 - 2}{2a_n} \;\geq\; 0,$$

using $a_n^2 \geq 2$ from (a) and $a_n > 0$. So (a_n) is decreasing in the sense of (D3).

It is bounded below by 0, since every $a_n > 0$. By monotone convergence (F7), (a_n) converges; call the limit L.

(c) Each $a_n \geq 0$, so $L \geq 0$ by (F4); and in fact $a_n^2 \geq 2$ gives $a_n \geq 2^{1/2} > 0$ by the monotonicity in (F11), so $L \geq 2^{1/2}$ by (F4) again. In particular $L \neq 0$.

Now pass to the limit in the recursion. The sequence $(a_{n+1})_{n\geq 1}$ is just (a_n) with its first term removed, so it also converges to L. By the limit laws (F3), using $L \neq 0$ and $a_n \neq 0$,

$$L \;=\; \lim_{n\to\infty} a_{n+1} \;=\; \lim_{n\to\infty} \frac{1}{2}\Big(a_n + \frac{2}{a_n}\Big) \;=\; \frac{1}{2}\Big(L + \frac{2}{L}\Big).$$

Multiplying by $2L$ gives $2L^2 = L^2 + 2$, that is $L^2 = 2$. Since $L > 0$ and $2^{1/2}$ is by (F11) the only non-negative real whose square is 2, we get $L = 2^{1/2}$.

(d) The identity is $(*)$, proved in (a). Write $e_n := a_n^2 - 2 \geq 0$ for the error in the square. Then $(*)$ says

$$e_{n+1} \;=\; \frac{e_n^2}{4a_n^2} \;\leq\; \frac{e_n^2}{8},$$

since $a_n^2 \geq 2$. So each error is at most one eighth of the *square* of the previous one. Squaring a small number roughly doubles the number of zeros standing

after its decimal point, so an error of size 10^{-k} is replaced by one of size about 10^{-2k}, and the count of correct digits roughly doubles at every step.

The three listed values bear this out. Starting from $e_1 = 2$ the bound gives $e_2 \le 2^2/8 = 5 \times 10^{-1}$, and the true value is 2.5×10^{-1}; starting from that, the bound gives $e_3 \le (2.5 \times 10^{-1})^2/8 \approx 7.8 \times 10^{-3}$, and the true value is 6.9×10^{-3}. Two further steps of the same recursion bring the error below 10^{-11}, so five terms already pin down $2^{1/2}$ to eleven decimal places. Compare this with a sequence such as $1/n$, where one must go ten times as far to gain a single extra digit.

Question (2). For a sequence $(a_n)_{n=1}^{\infty}$ of reals define its *averages*

$$\sigma_n \;:=\; \frac{a_1 + a_2 + \cdots + a_n}{n} \qquad (n \ge 1).$$

(a) Show that if $a_n \to L$ then $\sigma_n \to L$.

(b) Show that the converse fails, by giving a divergent sequence (a_n) for which (σ_n) converges.

Why this is interesting. Averaging smooths a sequence out, and part (b) shows it can smooth away a genuine oscillation. The sequence $(-1)^n$ does not converge, yet its averages converge to 0. One may therefore declare 0 to be the "average value" of a divergent sequence, and this is the starting point of summability theory, which assigns values to divergent series in a consistent way. Part (a) is the consistency requirement that makes such a declaration reasonable, since the new notion of limit must agree with the old one whenever the old one applies. The proof of (a) is also a model of a useful technique, splitting a sum into an early part that is fixed but possibly large, and a late part that is small.

(a) Suppose $a_n \to L$ and let $\varepsilon > 0$.

Step 1: control the late terms. By (D1) there is N_1 with

$$|a_n - L| \;\le\; \frac{\varepsilon}{2} \qquad \text{for all } n \ge N_1.$$

Step 2: split the sum. For $n \ge N_1$, note $\sigma_n - L = \frac{1}{n}\sum_{k=1}^{n}(a_k - L)$, and split this sum at N_1:

$$|\sigma_n - L| \;\le\; \frac{1}{n}\sum_{k=1}^{N_1-1} |a_k - L| \;+\; \frac{1}{n}\sum_{k=N_1}^{n} |a_k - L|,$$

by (F1) applied repeatedly. Call the two pieces P_n and Q_n.

Step 3: the late piece. Each of the at most n terms of the second sum is $\leq \varepsilon/2$, so

$$Q_n \;\leq\; \frac{1}{n} \cdot n \cdot \frac{\varepsilon}{2} \;=\; \frac{\varepsilon}{2}.$$

Step 4: the early piece. The quantity $C := \sum_{k=1}^{N_1-1} |a_k - L|$ is a fixed real number: it does not depend on n, being a sum of finitely many terms chosen once and for all. So $P_n = C/n$. By (F10) with $q := 1$ we have $1/n \to 0$, hence $C/n \to 0$ by (F3), and therefore there is N_2 with $P_n \leq \varepsilon/2$ for all $n \geq N_2$.

Conclusion. For $n \geq \max(N_1, N_2)$ we get $|\sigma_n - L| \leq P_n + Q_n \leq \varepsilon$. As $\varepsilon > 0$ was arbitrary, $\sigma_n \to L$.

(b) Take $a_n := (-1)^n$.

(a_n) *diverges.* Suppose $a_n \to M$. Applying (D1) with $\varepsilon := 1/2$ gives N with $|a_n - M| \leq 1/2$ for all $n \geq N$. Taking $n := 2N$ (even) and $n := 2N + 1$ (odd), both $\geq N$, and using (F1),

$$2 \;=\; |1 - (-1)| \;\leq\; |1 - M| + |M - (-1)| \;\leq\; \tfrac{1}{2} + \tfrac{1}{2} \;=\; 1,$$

which is false. So no such M exists.

(σ_n) *converges.* The partial sums of $(-1)^n$ are -1 for odd n and 0 for even n: indeed consecutive terms cancel in pairs. Hence

$$\sigma_n \;=\; \begin{cases} 0, & n \text{ even}, \\ -1/n, & n \text{ odd}, \end{cases}$$

so $|\sigma_n| \leq 1/n$ for every n. Given $\varepsilon > 0$, (F2) supplies a positive integer N with $1/\varepsilon \leq N$, and then $|\sigma_n - 0| \leq 1/n \leq 1/N \leq \varepsilon$ for all $n \geq N$. So $\sigma_n \to 0$.

The first few values are

$$\sigma_1 = -1, \quad \sigma_2 = 0, \quad \sigma_3 = -\tfrac{1}{3}, \quad \sigma_4 = 0, \quad \sigma_5 = -\tfrac{1}{5}, \quad \sigma_6 = 0, \; \ldots$$

Question (3). Let $(a_n)_{n=1}^{\infty}$ be a sequence of *positive* reals such that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} \;=\; r$$

exists, with $0 \leq r < 1$.

(a) Show that $a_n \to 0$.

(b) Deduce that $\lim_{n\to\infty} n^{10}/2^n = 0$ and $\lim_{n\to\infty} 2^n/n! = 0$.

Why this is interesting. This is the cleanest explanation of the hierarchy "polynomial beats nothing, exponential beats polynomial, factorial beats exponential", which is used constantly afterwards and in the analysis of algorithms. It is also a warning about numerical evidence. The sequence $n^{10}/2^n$ of part (b) tends to 0, but it first climbs to about 1.8×10^7 at $n = 14$ before collapsing, so an experiment that stopped at $n = 20$ would suggest exactly the wrong answer. The ratio sees past that hump, because it detects the trend rather than the size.

(a) Choose a real s with $r < s < 1$; for instance $s := (r+1)/2$, which satisfies $r < s < 1$ because $r < 1$.

Applying (D1) to the sequence of ratios with $\varepsilon := s - r > 0$, there is $N \geq 1$ with $|a_{n+1}/a_n - r| \leq s - r$ for all $n \geq N$, hence

$$\frac{a_{n+1}}{a_n} \;\leq\; r + (s-r) \;=\; s \qquad (n \geq N).$$

Since $a_n > 0$ this says $a_{n+1} \leq s\, a_n$ for $n \geq N$. By induction on n,

$$0 \;<\; a_n \;\leq\; s^{\,n-N} a_N \qquad (n \geq N):$$

it holds at $n = N$, and if it holds at n then $a_{n+1} \leq s a_n \leq s^{\,n+1-N} a_N$.

Write $A := a_N s^{-N} > 0$, so that $a_n \leq A s^n$ for $n \geq N$. Now $s^n \to 0$: indeed $\sum_{k=0}^{\infty} s^k$ converges by (F13) since $0 \leq s < 1$, so its terms tend to 0 by the zero test (F17). Hence $A s^n \to 0$ by (F3), and $0 \leq a_n \leq A s^n$ gives $a_n \to 0$ by the squeeze test (F5).

(b) *The first sequence.* Put $b_n := n^{10}/2^n > 0$. Then

$$\frac{b_{n+1}}{b_n} \;=\; \frac{(n+1)^{10}}{2^{n+1}} \cdot \frac{2^n}{n^{10}} \;=\; \frac{1}{2}\Big(1 + \frac{1}{n}\Big)^{10}.$$

By (F10), $1/n \to 0$, so $1 + 1/n \to 1$ by (F3); applying the product law in (F3) nine more times gives $(1 + 1/n)^{10} \to 1$, and once more $b_{n+1}/b_n \to 1/2$. Since $0 \leq 1/2 < 1$, part (a) gives $b_n \to 0$.

The second sequence. Put $c_n := 2^n/n! > 0$. Then

$$\frac{c_{n+1}}{c_n} \;=\; \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} \;=\; \frac{2}{n+1}.$$

Given $\varepsilon > 0$, (F2) supplies a positive integer N with $2/\varepsilon \leq N$, and then $|2/(n+1) - 0| = 2/(n+1) \leq 2/N \leq \varepsilon$ for $n \geq N$. So the ratios converge to $r := 0 < 1$, and part (a) gives $c_n \to 0$.

Question (4). Call a sequence $(a_n)_{n=1}^{\infty}$ of reals *k-contractive*, where $0 < k < 1$ is a real constant, if

$$|a_{n+1} - a_n| \;\le\; k\,|a_n - a_{n-1}| \qquad \text{for all } n \ge 2.$$

(a) Show that every k-contractive sequence converges. *(Hint: show first that $|a_{n+1} - a_n| \le k^{n-1}|a_2 - a_1|$, and then estimate $|a_j - a_i|$ for $j > i$ by a geometric sum.)*

(b) Let $a_1 := 1$ and $a_{n+1} := 1/(1 + a_n)$. Show that $\frac{1}{2} \le a_n \le 1$ for every n, that (a_n) is $\frac{4}{9}$-contractive, and find its limit.

Why this is interesting. Monotone convergence is the usual tool for proving that a recursively defined sequence converges, but it is useless here, because the sequence in (b) oscillates, jumping above and below its limit at every step. The contractive condition is the right substitute, and it needs neither monotonicity nor a known bound. It is also the engine of the fixed-point iterations used to solve equations numerically, since part (a) does not merely assert convergence but bounds the error at every stage, which is what makes such an iteration usable.

(a) Let (a_n) be k-contractive and put $d := |a_2 - a_1|$.

Step 1: the differences shrink geometrically. We claim $|a_{n+1} - a_n| \le k^{n-1}d$ for every $n \ge 1$. At $n = 1$ this reads $|a_2 - a_1| \le d$, which is an equality. If it holds at n, then

$$|a_{n+2} - a_{n+1}| \;\le\; k\,|a_{n+1} - a_n| \;\le\; k \cdot k^{n-1}d \;=\; k^n d,$$

which is the claim at $n + 1$. Induction completes the proof.

Step 2: the sequence is Cauchy. Let $j > i \ge 1$. Writing $a_j - a_i$ as the telescoping sum $\sum_{n=i}^{j-1}(a_{n+1} - a_n)$ and using (F1),

$$|a_j - a_i| \;\le\; \sum_{n=i}^{j-1} |a_{n+1} - a_n| \;\le\; \sum_{n=i}^{j-1} k^{n-1}d \;=\; k^{i-1}d \sum_{t=0}^{j-i-1} k^t \;\le\; k^{i-1}d \cdot \frac{1}{1-k},$$

the last step because $\sum_{t=0}^{T} k^t = (1 - k^{T+1})/(1 - k) \le 1/(1 - k)$ by (F13), using $0 < k < 1$.

Now $k^{i-1} \to 0$ as $i \to \infty$: the series $\sum_{t\ge 0} k^t$ converges by (F13), so its terms tend to 0 by (F17). Hence, given $\varepsilon > 0$, there is N with $k^{i-1}d/(1 - k) \le \varepsilon$ for all $i \ge N$, and then $|a_j - a_i| \le \varepsilon$ for all $j > i \ge N$. The same bound with the roles of i and j exchanged covers $i > j$, and the case $i = j$ is trivial. So (a_n) is Cauchy, and by (F6) it converges.

(b) *The bounds.* We show $\frac{1}{2} \le a_n \le 1$ by induction. It holds at $n = 1$, where

$a_1 = 1$. If $\frac{1}{2} \le a_n \le 1$, then $\frac{3}{2} \le 1 + a_n \le 2$, so

$$\tfrac{1}{2} \;\le\; \frac{1}{1+a_n} \;\le\; \tfrac{2}{3} \;\le\; 1,$$

which is the claim for a_{n+1}.

The contraction. For $n \ge 2$, a common denominator gives

$$a_{n+1} - a_n \;=\; \frac{1}{1+a_n} - \frac{1}{1+a_{n-1}} \;=\; \frac{a_{n-1} - a_n}{(1+a_n)(1+a_{n-1})},$$

so by (F1),

$$|a_{n+1} - a_n| \;=\; \frac{|a_n - a_{n-1}|}{(1+a_n)(1+a_{n-1})} \;\le\; \frac{|a_n - a_{n-1}|}{\frac{3}{2}\cdot\frac{3}{2}} \;=\; \tfrac{4}{9}\,|a_n - a_{n-1}|,$$

using $1 + a_n \ge \frac{3}{2}$ and $1 + a_{n-1} \ge \frac{3}{2}$. So (a_n) is $\frac{4}{9}$-contractive, and by (a) it converges; call the limit L.

The limit. Each $a_n \ge \frac{1}{2}$, so $L \ge \frac{1}{2}$ by (F4); in particular $1 + L \ne 0$ and every $1 + a_n \ne 0$. The sequence $(a_{n+1})_{n\ge1}$ converges to L as well, so by (F3),

$$L \;=\; \lim_{n\to\infty} a_{n+1} \;=\; \lim_{n\to\infty} \frac{1}{1+a_n} \;=\; \frac{1}{1+L}.$$

Hence $L(1 + L) = 1$, i.e. $L^2 + L - 1 = 0$, so $(2L + 1)^2 = 5$ and therefore $2L + 1 = 5^{1/2}$, the positive square root being the relevant one since $L \ge \frac{1}{2}$ makes $2L + 1 \ge 2 > 0$. Thus

$$L \;=\; \frac{5^{1/2} - 1}{2} \;=\; 0.6180339\ldots$$

Remark. The terms are $1, \frac{1}{2}, \frac{2}{3}, \frac{3}{5}, \frac{5}{8}, \frac{8}{13}, \frac{13}{21}, \ldots$: numerator and denominator are consecutive Fibonacci numbers, since $a_n = F_n/F_{n+1}$ gives $a_{n+1} = F_{n+1}/(F_{n+1} + F_n) = F_{n+1}/F_{n+2}$. So the limit just computed says that the ratio of consecutive Fibonacci numbers tends to $(5^{1/2} - 1)/2$, the reciprocal of the golden ratio; already $55/89 = 0.61797\ldots$ agrees with it to four decimal places.

Question (5). Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be bounded sequences of reals.

(a) Show that $\limsup_{n\to\infty}(a_n + b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$.

(b) Show that if (b_n) converges, then equality holds:

$$\limsup_{n\to\infty}(a_n + b_n) \;=\; \limsup_{n\to\infty} a_n + \lim_{n\to\infty} b_n.$$

Why this is interesting. Of all the operations in the subject, lim sup is the one students most often treat as if it were a limit, and the first casualty is additivity. It fails, and badly: $a_n := (-1)^n$ and $b_n := (-1)^{n+1}$ each have lim sup equal to 1, yet $a_n + b_n = 0$ for every n, so the sum has lim sup equal to 0 rather than 2. The reason is visible in the proof of part (a): $\limsup a_n$ and $\limsup b_n$ may be attained along different sets of indices, and the sum cannot be large at both sets at once. Part (b) identifies exactly what repairs the inequality, namely that one of the two sequences actually converges, so that it has nothing left to oscillate against.

Write $A := \limsup_{n\to\infty} a_n$ and $B := \limsup_{n\to\infty} b_n$. By (D2) the hypothesis supplies reals $M_1, M_2 \geq 0$ with $|a_n| \leq M_1$ and $|b_n| \leq M_2$ for all n. Hence each of the sets $\{a_n : n \geq N\}$ is non-empty and bounded, so by (F8) every a_N^+ of (D6) is a real number with $-M_1 \leq a_N^+ \leq M_1$. The quantity A is the infimum of these, a non-empty set of reals all lying in $[-M_1, M_1]$, so A is real as well, and likewise B. Since $|a_n + b_n| \leq M_1 + M_2$ by (F1), the same applies to the sequence $(a_n + b_n)$.

(a) Fix $N \geq 1$. For every $n \geq N$ we have, by (F8),

$$a_n + b_n \;\leq\; a_N^+ + b_N^+.$$

So $a_N^+ + b_N^+$ is an upper bound for $\{a_n + b_n : n \geq N\}$, and (F8) gives

$$(a+b)_N^+ \;\leq\; a_N^+ + b_N^+,$$

writing $(a+b)_N^+$ for the quantity of (D6) formed from the sequence $(a_n + b_n)$.

Now $A \leq a_N^+$ and $B \leq b_N^+$, again by (F8), since A and B are infima of sets containing a_N^+ and b_N^+ respectively. We cannot simply combine these, since they point the wrong way; instead we let N grow. Let $\varepsilon > 0$. By (F8) applied to the infimum defining A there is N_1 with $a_{N_1}^+ < A + \varepsilon/2$, and similarly N_2 with $b_{N_2}^+ < B + \varepsilon/2$. Put $N_0 := \max(N_1, N_2)$. The quantities a_N^+ decrease as N increases, because the set $\{a_n : n \geq N\}$ shrinks and so its supremum cannot grow; hence

$$(a+b)_{N_0}^+ \;\leq\; a_{N_0}^+ + b_{N_0}^+ \;\leq\; a_{N_1}^+ + b_{N_2}^+ \;<\; A + B + \varepsilon.$$

Since $\limsup_{n\to\infty}(a_n + b_n)$ is the infimum of all the $(a+b)_N^+$, it is $\leq (a+b)_{N_0}^+ < A + B + \varepsilon$ by (F8). As $\varepsilon > 0$ was arbitrary, $\limsup_{n\to\infty}(a_n + b_n) \leq A + B$.

(b) Suppose $b_n \to \beta$. By (a) it suffices to prove the reverse inequality $\limsup_{n\to\infty}(a_n + b_n) \geq A + \beta$, together with $B = \beta$.

First, $B = \beta$. Let $\varepsilon > 0$ and take N with $|b_n - \beta| \leq \varepsilon$ for $n \geq N$. Then $\beta + \varepsilon$ is an upper bound for $\{b_n : n \geq N\}$ and $\beta - \varepsilon$ a lower bound, so $\beta - \varepsilon \leq b_N^+ \leq \beta + \varepsilon$

by (F8). Since B is the infimum of the b_N^+ and these all lie in $[\beta-\varepsilon, \beta+\varepsilon]$ for N large, and since they decrease, B lies in that interval too; as ε was arbitrary, $B = \beta$.

Now the reverse inequality. Apply part (a) to the two sequences $(a_n + b_n)$ and $(-b_n)$, whose sum is (a_n):

$$A \;=\; \limsup_{n\to\infty} a_n \;\le\; \limsup_{n\to\infty}(a_n+b_n) + \limsup_{n\to\infty}(-b_n).$$

Since $-b_n \to -\beta$, the paragraph above applied to $(-b_n)$ gives $\limsup_{n\to\infty}(-b_n) = -\beta$. Hence $A \le \limsup_{n\to\infty}(a_n + b_n) - \beta$, that is $\limsup_{n\to\infty}(a_n + b_n) \ge A + \beta$. Combined with (a), equality holds.

Question (6). List the rationals in $[0,1]$ with denominator at most n, for $n = 1$, then $n = 2$, then $n = 3$, and so on, forming the sequence

$$\underbrace{\tfrac{0}{1},\ \tfrac{1}{1}}_{n=1},\ \underbrace{\tfrac{0}{2},\ \tfrac{1}{2},\ \tfrac{2}{2}}_{n=2},\ \underbrace{\tfrac{0}{3},\ \tfrac{1}{3},\ \tfrac{2}{3},\ \tfrac{3}{3}}_{n=3},\ \underbrace{\tfrac{0}{4},\ \tfrac{1}{4},\ \tfrac{2}{4},\ \tfrac{3}{4},\ \tfrac{4}{4}}_{n=4},\ \ \dots$$

Call the resulting sequence $(x_1, x_2, x_3, \dots)$; thus $x_1 = 0$, $x_2 = 1$, $x_3 = 0$, $x_4 = \frac{1}{2}$, $x_5 = 1$, $x_6 = 0$, and so on. Show that every $x \in [0,1]$ is a limit point of (x_j).

Why this is interesting. A convergent sequence has exactly one limit point, and it is tempting to carry that picture over to sequences in general. This example demolishes the picture as thoroughly as possible, with a single explicit sequence of rationals whose limit points fill an entire interval. Everything in the construction is concrete — one can write down the first twenty terms — yet the set of limit points is uncountable. It is also a good reminder that "limit point" is a much weaker notion than "limit": the sequence gets close to $\frac{1}{\sqrt{2}}$ infinitely often without ever going there.

First note what the construction guarantees. The block indexed by n consists of the $n+1$ numbers $0/n, 1/n, \dots, n/n$, and the blocks before it, indexed by $1, 2, \dots, n-1$, occupy $2 + 3 + \cdots + n$ places in all. So every one of the listed values sits at a definite and finite index, and we may assert the following.

> *for every $n \ge 1$ and every integer k with $0 \le k \le n$, the value k/n occurs in the sequence, and it occurs at arbitrarily late indices.* (∗)

The last clause holds because k/n also equals $2k/2n$, then $3k/3n$, and so on. Each of these is one of the values listed in a later block, since $0 \le 2k \le 2n$, $0 \le 3k \le 3n$, and so forth, so the same number reappears arbitrarily far along the sequence.

Now let $x \in [0,1]$. By (D5) we must show that for every $\varepsilon > 0$ and every $N \ge 1$

there is some $j \geq N$ with $|x_j - x| \leq \varepsilon$.

So let $\varepsilon > 0$ and $N \geq 1$ be given. By (F2) choose a positive integer n with $1/\varepsilon \leq n$, so $1/n \leq \varepsilon$. By (F12) put $k := \lfloor nx \rfloor$, so that $k \leq nx < k+1$ and hence, dividing by $n > 0$,

$$\frac{k}{n} \;\leq\; x \;<\; \frac{k+1}{n}, \qquad \text{so} \qquad \left|x - \frac{k}{n}\right| \;<\; \frac{1}{n} \;\leq\; \varepsilon.$$

Moreover $0 \leq k \leq n$: from $x \geq 0$ we get $nx \geq 0$ and so $k \geq 0$ (if $k \leq -1$ then $nx < k+1 \leq 0$, a contradiction); and from $x \leq 1$ we get $k \leq nx \leq n$.

So k/n is one of the listed values. By $(*)$ it occurs at arbitrarily late indices, so there is $j \geq N$ with $x_j = k/n$, and then $|x_j - x| < \varepsilon$. Hence x is a limit point.

Part B. Series

Question (7). *(Cauchy's condensation test.)* Let $(a_n)_{n=1}^{\infty}$ be non-negative and decreasing.

(a) Show that for every $k \geq 0$,

$$\sum_{n=1}^{2^k} a_n \;\geq\; \tfrac{1}{2} \sum_{j=0}^{k} 2^j a_{2^j} \qquad \text{and} \qquad \sum_{n=1}^{2^{k+1}-1} a_n \;\leq\; \sum_{j=0}^{k} 2^j a_{2^j}.$$

(b) Deduce that $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{j=0}^{\infty} 2^j a_{2^j}$ converges.

Why this is interesting. The test replaces a series by one with exponentially sparser terms, and the two stand or fall together. That is a remarkable economy, since the fate of the whole series is then decided by the terms at the indices $1, 2, 4, 8, 16, \ldots$ alone. The mechanism is worth understanding — the terms are grouped into blocks of lengths $1, 2, 4, 8, \ldots$, and because the sequence is decreasing each block is pinned between 2^j copies of its first term and 2^j copies of its last. The pay-off comes when it is applied to $a_n := 1/n^p$: the condensed series is then geometric with ratio 2^{1-p}, which settles the convergence of $\sum 1/n^p$ for every p at one stroke, and locates the threshold at $p = 1$.

Throughout, write $S_n := \sum_{i=1}^{n} a_i$ and $T_k := \sum_{j=0}^{k} 2^j a_{2^j}$. All terms are non-negative, so both (S_n) and (T_k) are increasing.

(a) *The upper estimate.* Group the indices $1 \le n \le 2^{k+1}-1$ into the blocks

$$\{1\},\ \{2,3\},\ \{4,5,6,7\},\ \ldots,\ \{2^k,\ldots,2^{k+1}-1\},$$

the jth of which is $\{2^j,\ldots,2^{j+1}-1\}$ and has exactly 2^j members. Every n in that block satisfies $n \ge 2^j$, so $a_n \le a_{2^j}$ because (a_n) is decreasing. Hence the block contributes at most $2^j a_{2^j}$, and summing over $j = 0,\ldots,k$,

$$S_{2^{k+1}-1} \ \le\ \sum_{j=0}^{k} 2^j a_{2^j} \ =\ T_k.$$

The lower estimate. Now group $2 \le n \le 2^k$ into the blocks $\{2\},\{3,4\},\{5,\ldots,8\},\ldots,\{2^{j-1}+1,\ldots,2^j\},\ldots$ for $j = 1,\ldots,k$; the jth has 2^{j-1} members, each $n \le 2^j$, so $a_n \ge a_{2^j}$ and the block contributes at least $2^{j-1}a_{2^j}$. Adding the term $n = 1$ separately,

$$S_{2^k} \ \ge\ a_1 + \sum_{j=1}^{k} 2^{j-1} a_{2^j} \ \ge\ \tfrac{1}{2}a_1 + \sum_{j=1}^{k} \tfrac{1}{2}\cdot 2^j a_{2^j} \ =\ \tfrac{1}{2}T_k,$$

using $a_1 \ge \frac{1}{2}a_1$, which holds as $a_1 \ge 0$.

(b) Both series have non-negative terms, so by (F14) each converges exactly when its partial sums are bounded above.

If $\sum 2^j a_{2^j}$ converges, its partial sums T_k are bounded above, say $T_k \le C$ for all k. Let $n \ge 1$ be given; we claim there is a k with $n \le 2^{k+1}-1$. By (F2) choose k with $k+1 \ge n$. An easy induction shows $2^m \ge m+1$ for every $m \ge 0$, since $2^0 = 1$ and $2^{m+1} = 2\cdot 2^m \ge 2m+2 \ge m+2$; applying it with $m := k+1$ gives $2^{k+1} \ge k+2 \ge n+1$, which is the claim.

All the terms are non-negative, so (S_n) is increasing and $S_n \le S_{2^{k+1}-1}$. Combining this with the upper estimate of part (a),

$$S_n \ \le\ S_{2^{k+1}-1} \ \le\ T_k \ \le\ C.$$

So the S_n are bounded above and $\sum a_n$ converges.

If $\sum a_n$ converges, its partial sums are bounded above, say $S_n \le C$. Then by (a), $T_k \le 2S_{2^k} \le 2C$ for every k, so the T_k are bounded above and $\sum 2^j a_{2^j}$ converges.

Remark. The test is exactly as strong as the decreasing hypothesis. Without it the blocks cannot be pinned between their endpoints, and the conclusion fails. One can rearrange the terms of a convergent series of non-negative terms into a non-monotone sequence whose condensation diverges.

Question (8). Let $(a_n)_{n=1}^{\infty}$ be a sequence of non-negative reals.

(a) Show that if $\sum_{n=1}^{\infty} a_n$ converges then so does $\sum_{n=1}^{\infty} a_n^2$.

(b) Show that the converse fails.

> ***Why this is interesting.*** A convergent series of non-negative terms has terms that are eventually small, and squaring a small number makes it much smaller; part (a) turns this into a proof. The real content is part (b), which shows the implication is strictly one-way, so "$\sum a_n^2$ converges" is a genuinely weaker condition. The example $a_n = 1/n$ witnesses that failure in the sharpest possible way: $\sum 1/n$ diverges while $\sum 1/n^2$ converges, so one single sequence separates the two conditions.

(a) Suppose $\sum a_n$ converges. By the zero test (F17), $a_n \to 0$, so applying (D1) with $\varepsilon := 1$ there is N with $|a_n - 0| \le 1$, i.e. $0 \le a_n \le 1$, for all $n \ge N$.

For such n, multiplying $a_n \le 1$ by $a_n \ge 0$ gives

$$0 \;\le\; a_n^2 \;\le\; a_n \qquad (n \ge N).$$

The series $\sum_{n \ge N} a_n$ converges, being $\sum a_n$ with finitely many terms removed. By the comparison test (F15), $\sum_{n \ge N} a_n^2$ converges; and restoring the finitely many terms $a_1^2, \ldots, a_{N-1}^2$ leaves it convergent. So $\sum a_n^2$ converges.

(b) Take $a_n := 1/n$.

$\sum a_n^2$ converges. For $n \ge 2$, $n(n-1) \le n^2$ gives $1/n^2 \le 1/(n(n-1)) = 1/(n-1) - 1/n$, whose partial sums from 2 to M telescope to $1 - 1/M \le 1$. So the partial sums of $\sum_{n \ge 2} 1/n^2$ are bounded above by 1, and (F14) gives convergence; adding the term $n = 1$ keeps it convergent.

$\sum a_n$ diverges. Suppose it converged, with partial sums H_M. For every $m \ge 1$,

$$H_{2m} - H_m \;=\; \sum_{i=m+1}^{2m} \frac{1}{i} \;\ge\; m \cdot \frac{1}{2m} \;=\; \frac{1}{2},$$

since the sum has m terms, each at least $1/(2m)$. But a convergent sequence is Cauchy by (F6), so taking $\varepsilon := 1/4$ there would be N with $|H_j - H_i| \le 1/4$ for all $i, j \ge N$; taking $i := N$ and $j := 2N$ gives $1/2 \le 1/4$, which is false.

So the converse of (a) fails, since $\sum a_n^2$ converges while $\sum a_n$ does not.

Question (9). Say that a series $\sum_{k=1}^{\infty} b_k$ is obtained from $\sum_{n=1}^{\infty} a_n$ by *grouping* if there are indices $0 = m_0 < m_1 < m_2 < \cdots$ with

$$b_k \;:=\; a_{m_{k-1}+1} + a_{m_{k-1}+2} + \cdots + a_{m_k} \qquad (k \ge 1),$$

so that $\sum b_k$ is $\sum a_n$ with its terms collected into consecutive groups, no term being moved, repeated or omitted.

(a) Show that if $\sum a_n$ converges to L, then every grouping of it converges to L.

(b) Show that a grouping can converge while the original series diverges.

Why this is interesting. "Regrouping terms cannot change a sum" is a habit carried over from finite arithmetic, and part (b) shows the price of carrying it too far. The grouping $(1-1)+(1-1)+\cdots$ produces 0 from a series that has no sum at all. This is Grandi's series, which caused genuine confusion in the eighteenth century, and part (a) is the resolution. Grouping is safe in one direction only, because the grouped partial sums are nothing but *some* of the original partial sums, namely those taken at the ends of the groups; and a subsequence of a divergent sequence may perfectly well converge, so the grouped series can settle down while the original one still oscillates.

Write $S_n := \sum_{i=1}^{n} a_i$ and $B_k := \sum_{j=1}^{k} b_j$ for the two sequences of partial sums. Everything rests on one observation. Adding the first k groups uses each of $a_1, \ldots, a_{m_k}$ exactly once, so

$$B_k \;=\; a_1 + \cdots + a_{m_k} \;=\; S_{m_k}.$$

That is, the grouped partial sums are exactly the original partial sums read off at the indices $m_1 < m_2 < m_3 < \cdots$. Since these indices are strictly increasing, (B_k) is a subsequence of (S_n) in the sense of (D4).

(a) If $\sum a_n$ converges to L, then $S_n \to L$ by (D7). By (F9), every subsequence of a convergent sequence converges to the same limit, so $B_k \to L$. Hence the grouped series converges, with the same sum L.

Note that the argument needs nothing about the sizes of the groups. However long or irregular they are, the grouped partial sums are drawn from a sequence that has already settled down.

(b) Take $a_n := (-1)^{n+1}$, so the series is $1 - 1 + 1 - 1 + \cdots$, and group in consecutive pairs by putting $m_k := 2k$, so that

$$b_k \;=\; a_{2k-1} + a_{2k} \;=\; 1 + (-1) \;=\; 0.$$

The grouped series is $0+0+0+\cdots$, whose partial sums are all 0, so it converges to 0.

But $\sum a_n$ diverges. Its terms satisfy $|a_n| = 1$ for every n, so $a_n \not\to 0$; indeed with $\varepsilon := \frac{1}{2}$ no N can satisfy $|a_n - 0| \le \frac{1}{2}$ for all $n \ge N$. By the zero test (F17) a convergent series has terms tending to 0, so this series cannot converge.

Seen through the observation above, there is no paradox. The original partial sums run $1, 0, 1, 0, \ldots$ and have no limit, while the grouping picks out the even-numbered ones, which are all 0.

Remark. Regrouping the same series differently gives a different answer. Taking $1 + (-1 + 1) + (-1 + 1) + \cdots$ picks out the odd-numbered partial sums instead, every one of which equals 1, so this grouping converges to 1. That two groupings of one series give 0 and 1 is precisely the signal that the ungrouped series has no sum for them to preserve.

Question (10). Let

$$S \;:=\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \;=\; 1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots,$$

and let T be the series built from the same terms by taking them *two positive, one negative* in turn:

$$T \;:=\; 1 + \tfrac{1}{3} - \tfrac{1}{2} \;+\; \tfrac{1}{5} + \tfrac{1}{7} - \tfrac{1}{4} \;+\; \tfrac{1}{9} + \tfrac{1}{11} - \tfrac{1}{6} \;+\; \cdots,$$

so that the mth group of three terms is $\frac{1}{4m-3} + \frac{1}{4m-1} - \frac{1}{2m}$.

(a) Verify that T uses every term of S exactly once, in a different order.

(b) Writing S_N and T_N for the Nth partial sums, show that

$$T_{3M} \;=\; S_{4M} + \tfrac{1}{2} S_{2M} \qquad (M \geq 1),$$

and deduce that T converges, with sum $\frac{3}{2}S$.

Why this is interesting. Addition is commutative, and it is very hard to believe that adding the same numbers in a different order can change the answer. For a conditionally convergent series it does. The explanation is that the sum of a series is the limit of its partial sums, and reordering changes which terms have been reached by stage N: here, taking two positive terms for every negative one keeps the partial sums permanently ahead, and the excess does not wash out. Riemann went further and showed that a conditionally convergent series can be rearranged to converge to *any* prescribed real number, or to diverge; this question is one step of that theorem, and the reason absolute convergence is the hypothesis one insists on whenever rearrangement is involved.

Throughout write $H_M := 1 + \frac{1}{2} + \cdots + \frac{1}{M}$.

(a) The terms of S are the positive terms $1/(2k-1)$ for $k \geq 1$, that is $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \ldots$, together with the negative terms $-1/(2j)$ for $j \geq 1$, that is $-\frac{1}{2}, -\frac{1}{4}, -\frac{1}{6}, \ldots$.

The positive terms of the mth group of T are $1/(4m-3)$ and $1/(4m-1)$. As m runs over $1, 2, 3, \ldots$, the index $4m-3$ runs over $1, 5, 9, 13, \ldots$ and $4m-1$ runs over $3, 7, 11, 15, \ldots$; together these are exactly the odd positive integers, each occurring once. So every positive term of S is used exactly once.

The negative term of the mth group is $-1/(2m)$, and as m runs over the positive integers these are exactly the negative terms of S, each once.

Hence T is a rearrangement of S: the same terms, reordered.

(b) *The identity.* Summing the mth group over $m = 1, \ldots, M$ splits T_{3M} into its positive and negative halves:

$$T_{3M} \;=\; \sum_{m=1}^{M} \Big(\frac{1}{4m-3} + \frac{1}{4m-1} \Big) \;-\; \sum_{m=1}^{M} \frac{1}{2m}.$$

By part (a) the first sum is $\sum 1/j$ over the odd j with $1 \leq j \leq 4M-1$. Subtracting the even j from all j, and writing each even j as $2i$ with $1 \leq i \leq 2M$,

$$\sum_{\substack{j \leq 4M \\ j \text{ odd}}} \frac{1}{j} \;=\; H_{4M} - \sum_{i=1}^{2M} \frac{1}{2i} \;=\; H_{4M} - \tfrac{1}{2} H_{2M},$$

while the second sum is $\frac{1}{2} H_M$. Hence

$$T_{3M} \;=\; H_{4M} - \tfrac{1}{2} H_{2M} - \tfrac{1}{2} H_M.$$

On the other side, split S_{2K} by parity. Its odd-indexed terms contribute $\sum 1/j$ over the odd $j \leq 2K$, which by the same subtraction is $H_{2K} - \frac{1}{2} H_K$; its even-indexed terms contribute $-\sum_{i=1}^{K} 1/(2i) = -\frac{1}{2} H_K$. Hence $S_{2K} = H_{2K} - H_K$, and so

$$S_{4M} + \tfrac{1}{2} S_{2M} \;=\; \big(H_{4M} - H_{2M} \big) + \tfrac{1}{2} \big(H_{2M} - H_M \big) \;=\; H_{4M} - \tfrac{1}{2} H_{2M} - \tfrac{1}{2} H_M,$$

which is the same expression. This proves the identity.

Convergence of S. The sequence $1/n$ is non-negative and decreasing, and tends to 0 by (F10); so by the alternating series test (F16) the series $\sum_{n \geq 1} (-1)^n / n$ converges, and hence so does S, which is its negative.

Passing to the limit. Both $(S_{4M})_{M\geq 1}$ and $(S_{2M})_{M\geq 1}$ are subsequences of the convergent sequence (S_N), so both converge to S by (F9). By the identity and the limit laws (F3),

$$\lim_{M\to\infty} T_{3M} \;=\; S + \tfrac{1}{2}S \;=\; \tfrac{3}{2}S.$$

The remaining partial sums. Reading off the next two terms of T,

$$T_{3M+1} = T_{3M} + \frac{1}{4M+1}, \qquad T_{3M+2} = T_{3M+1} + \frac{1}{4M+3}.$$

Both corrections tend to 0: given $\varepsilon > 0$, (F2) supplies a positive integer P with $1/\varepsilon \leq P$, and for $M \geq P$ both $1/(4M+1)$ and $1/(4M+3)$ are at most $1/M \leq 1/P \leq \varepsilon$. So by (F3) each of the three sequences (T_{3M}), (T_{3M+1}), (T_{3M+2}) converges to $\frac{3}{2}S$.

Finally every index N has one of the three forms, so given $\varepsilon > 0$ we may take N_0 beyond the thresholds for all three and conclude $|T_N - \frac{3}{2}S| \leq \varepsilon$ for $N \geq N_0$. Hence T converges, with sum $\frac{3}{2}S$.

Remark. The two sums really are different. Collecting the terms of S in consecutive pairs gives

$$S_{2M} \;=\; \sum_{m=1}^{M}\Big(\frac{1}{2m-1} - \frac{1}{2m}\Big),$$

in which every pair contributes a positive amount because $2m-1 < 2m$. Keeping only the first of them, $S_{2M} \geq 1 - \frac{1}{2} = \frac{1}{2}$ for every M, so $S \geq \frac{1}{2}$ by (F9) and (F4). Hence $S \neq 0$ and $\frac{3}{2}S \neq S$, which is to say that T and S have exactly the same terms and different sums. Numerically $S = 0.6931\ldots$ while $T = 1.0397\ldots$, and the drift shows up early. After six terms of each, $T_6 = \frac{389}{420} = 0.9262\ldots$ against $S_6 = 0.6167\ldots$

Remark. The series S is conditionally convergent in the sense of (D8), since it converges while $\sum_{n\geq 1} 1/n$ does not. That is exactly the room the rearrangement exploits. For an absolutely convergent series no such thing can happen, and every rearrangement has the same sum.

Question (11).

(a) Show that $\sum_{n=1}^{\infty} 9/10^n = 1$. (This is the statement that $0.999\ldots = 1$.)

(b) Let $d_1, d_2, \ldots, d_r$ be digits, not all zero, and let

$$x \;:=\; \sum_{k=0}^{\infty} \frac{D}{10^{r(k+1)}}, \qquad \text{where } D := d_1 d_2 \ldots d_r \text{ as an } r\text{-digit integer.}$$

Show that the series converges and that $x = D/(10^r - 1)$.

Why this is interesting. Every student has been told that $0.999\ldots = 1$ and most remain unconvinced, because the usual arguments move an infinite decimal around as if it were a finite one. Once a decimal expansion is *defined* as the sum of a series, there is nothing left to be uneasy about. The two symbols denote the same real number because the series has sum 1. Part (b) then explains the familiar trick of multiplying by 10^r and subtracting, by showing it is nothing but the geometric series formula, and it recovers facts from school arithmetic — that $1/7$ has period six, for instance — as statements about convergent series.

(a) The series is $\sum_{n=1}^{\infty} 9 \cdot (1/10)^n$. Factoring out one power,

$$\sum_{n=1}^{N} 9\Big(\frac{1}{10}\Big)^n \;=\; \frac{9}{10}\sum_{k=0}^{N-1}\Big(\frac{1}{10}\Big)^k \;=\; \frac{9}{10}\cdot\frac{1-(1/10)^N}{1-1/10} \;=\; 1-\Big(\frac{1}{10}\Big)^N,$$

using the finite geometric formula in (F13) with $r := 1/10$.

Now $(1/10)^N \to 0$: the series $\sum_{k\ge 0}(1/10)^k$ converges by (F13), so its terms tend to 0 by the zero test (F17). Hence by (F3) the partial sums converge to 1, i.e.

$$\sum_{n=1}^{\infty} \frac{9}{10^n} \;=\; 1.$$

(b) Write $\rho := 10^{-r}$, so $0 < \rho < 1$, and note

$$\frac{D}{10^{r(k+1)}} \;=\; D\,\rho^{\,k+1} \;=\; D\rho\cdot\rho^k.$$

So the series is $D\rho\sum_{k=0}^{\infty}\rho^k$, which converges by (F13) since $0 \le \rho < 1$, with

$$x \;=\; D\rho\cdot\frac{1}{1-\rho} \;=\; \frac{D\cdot 10^{-r}}{1-10^{-r}} \;=\; \frac{D}{10^r-1},$$

the last step on multiplying numerator and denominator by 10^r.

That x is the decimal $0.\overline{d_1\ldots d_r}$ is exactly what the series says, since the kth term places the block $d_1\ldots d_r$ in the decimal positions $rk+1$ through $r(k+1)$. Since D and $10^r - 1$ are integers with $10^r - 1 \neq 0$, the number x is rational.

Remark. Two instances are worth noting. With $r = 6$ and $D = 142857$ the formula gives $142857/999999 = 1/7$, which is why $1/7$ has period six; and with $r = 1$ and $D = 9$ it gives $9/9 = 1$, recovering part (a).

Remark. The two decimals $1.000\ldots$ and $0.999\ldots$ are different symbols naming the same real number, and this is the only way decimal notation fails to be unique. It happens exactly for the numbers with a terminating expansion, each of which has a second expansion ending in repeating 9s.

Part C. The real numbers

Question (12). For non-empty sets $A, B \subseteq \mathbb{R}$ write

$$A + B := \{\, a + b : a \in A,\ b \in B \,\}, \qquad A \cdot B := \{\, ab : a \in A,\ b \in B \,\}.$$

(a) Suppose A and B are bounded above. Show that $A + B$ is bounded above and $\sup(A + B) = \sup(A) + \sup(B)$.

(b) Show that the corresponding statement for products is false, by finding bounded A, B with $\sup(A \cdot B) \neq \sup(A)\sup(B)$.

Why this is interesting. Part (a) is one of the few places where a supremum behaves perfectly. It is additive, with no hypothesis beyond non-emptiness and boundedness. That makes part (b) all the more striking, since multiplication is in every other respect as well-behaved as addition. The reason is visible once one looks for it. A large product can come from two large negative numbers, which are nowhere near the suprema. The same phenomenon is behind the rule of thumb that multiplying a set by a negative number exchanges sup and inf — the same reason an inequality flips when multiplied by -1.

(a) Write $\alpha := \sup(A)$ and $\beta := \sup(B)$, real numbers by (F8).

Upper bound. For $a \in A$ and $b \in B$ we have $a \leq \alpha$ and $b \leq \beta$ by (F8), so $a + b \leq \alpha + \beta$. Hence $A + B$ is non-empty (both A and B are) and bounded above by $\alpha + \beta$, so $\sup(A + B)$ is real and

$$\sup(A + B) \ \leq\ \alpha + \beta$$

by (F8).

Reverse inequality. Let $\varepsilon > 0$. By (F8) there are $a \in A$ with $a > \alpha - \varepsilon/2$ and $b \in B$ with $b > \beta - \varepsilon/2$. Then $a + b \in A + B$ and

$$\sup(A + B) \ \geq\ a + b \ >\ \alpha + \beta - \varepsilon,$$

the first step by (F8). So $\sup(A + B) > \alpha + \beta - \varepsilon$ for every $\varepsilon > 0$. If we had $\sup(A + B) < \alpha + \beta$, then taking $\varepsilon := \alpha + \beta - \sup(A + B) > 0$ would give $\sup(A + B) > \sup(A + B)$, which is absurd. Hence $\sup(A + B) \geq \alpha + \beta$, and equality holds.

(b) Take $A = B := \{-2, 1\}$. Then $\sup(A) = \sup(B) = 1$, so $\sup(A)\sup(B) = 1$. But

$$A \cdot B \ =\ \{\,(-2)(-2),\ (-2)(1),\ (1)(-2),\ (1)(1)\,\} \ =\ \{\,4,\ -2,\ 1\,\},$$

whose supremum is 4. So $\sup(A \cdot B) = 4 \neq 1 = \sup(A)\sup(B)$.

The largest product arises from the two *smallest* elements, which the suprema know nothing about.

Remark. The failure is caused entirely by the negative numbers. If every element of A and of B is non-negative, then $\sup(A \cdot B) = \sup(A)\sup(B)$, by exactly the argument used in (a) with sums replaced by products.